

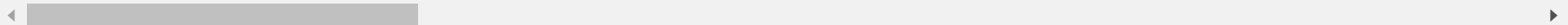
```
In [ ]: import pandas as pd
```

```
In [ ]: df = pd.read_excel("data/promoters_alignment_all.xlsx")
df.head()
```

```
Out[ ]:
```

	id	species	genus	family	order	RNAI_p
0	Aeromonas salmonicida_Aeromonadaceae...	Aeromonas salmonicida	Aeromonas	Aeromonadaceae	Aeromonadales	AGTATCAAACCCCTGCTCGAACCTGTTACCAAA
1	Aeromonas hydrophila_Aeromonadaceae...	Aeromonas hydrophila	Aeromonas	Aeromonadaceae	Aeromonadales	ATGGCGATTTGTCGTCCTTTTCGGGTTGGACTC
2	Aeromonas media_Aeromonadaceae_Aerom...	Aeromonas media	Aeromonas	Aeromonadaceae	Aeromonadales	TTTCAAGCTAATCCCTCTTTACGCCGTTACCGA
3	Aeromonas salmonicida_Aeromonadaceae...	Aeromonas salmonicida	Aeromonas	Aeromonadaceae	Aeromonadales	GTATCAAACCCCTGCTCGAACCTGTTACCAAA
4	Aeromonas salmonicida_Aeromonadaceae...	Aeromonas salmonicida	Aeromonas	Aeromonadaceae	Aeromonadales	GTATCAAACCCCTGCTCGAACCTGTTACCAAA

5 rows × 21 columns

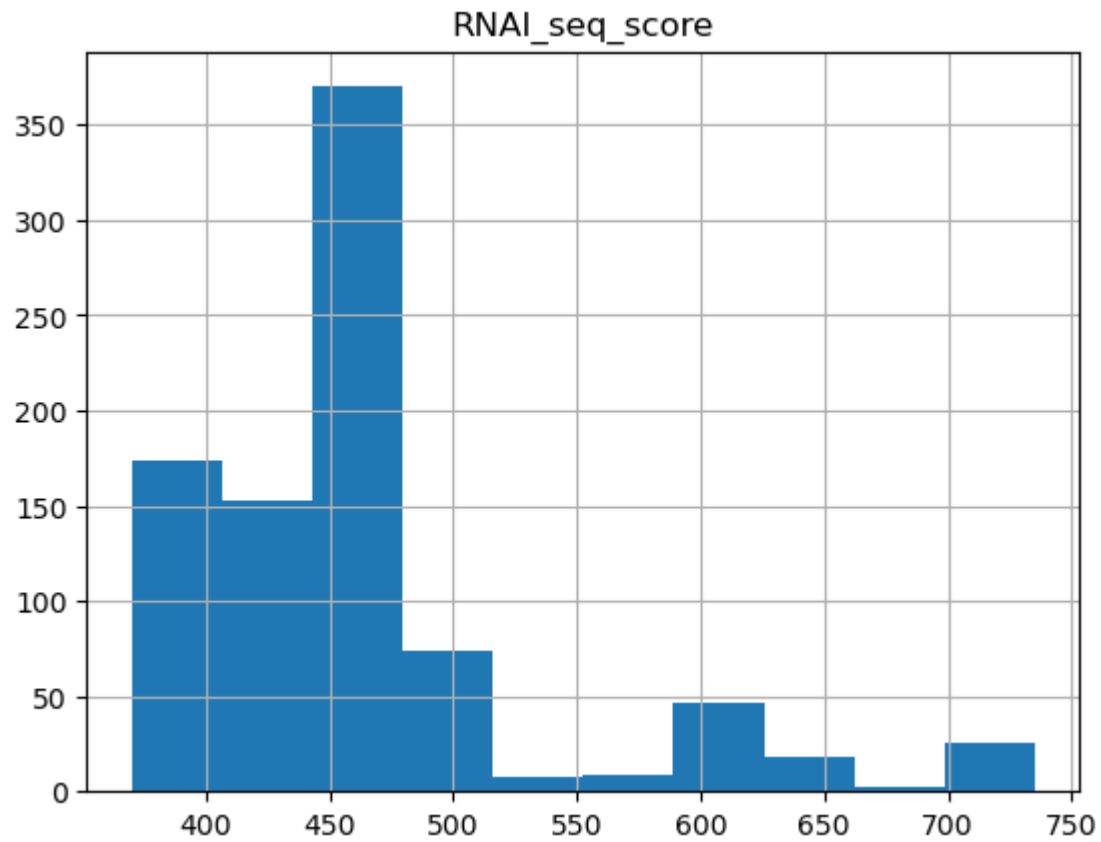


```
In [ ]: # Current null model (other strand) as a lower bound threshold for alignment:
puc19_alignment_score_threshold = {
    "RNAI_seq": 369.0,
    "RNAP_seq": 1272.0,
}

# Max score of alignment (as an upper bound threshold for alignment):
puc19_alignment_max_score = {
    "RNAI_seq": 735.0,
    "RNAP_seq": 2765.0,
}
```

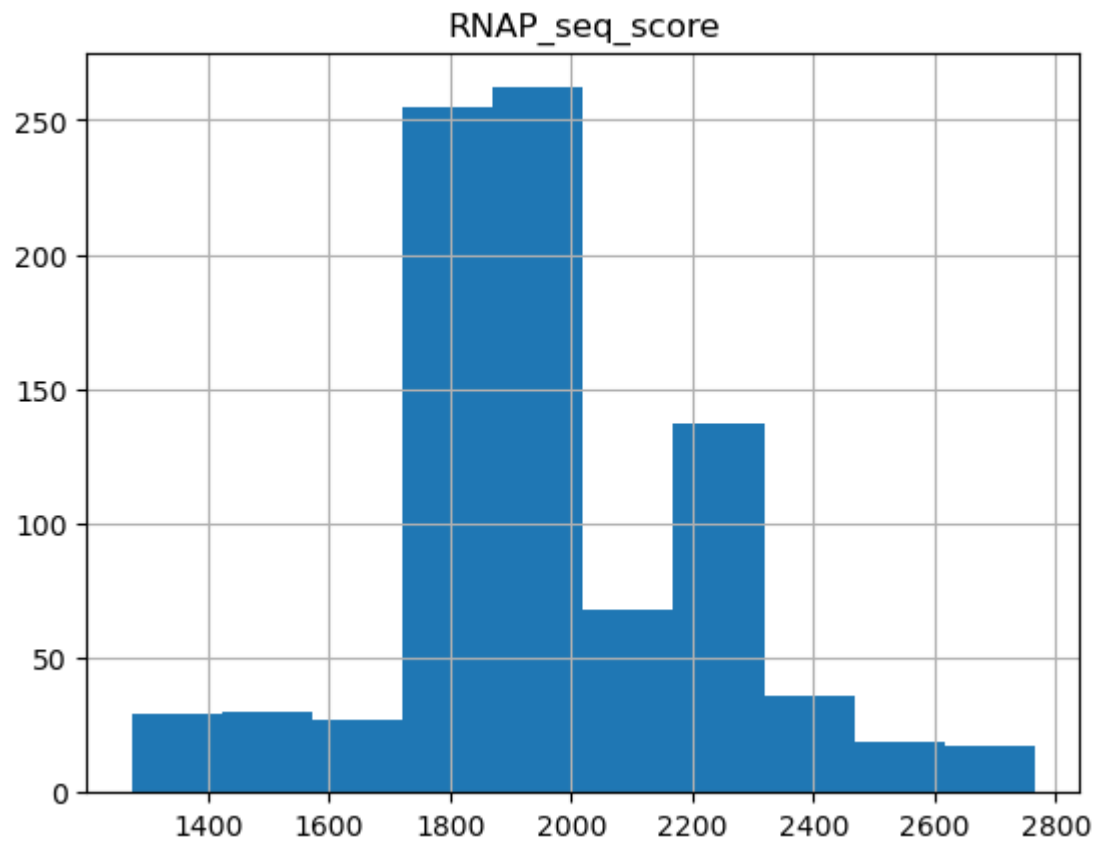
```
In [ ]: df.hist(column="RNAI_seq_score")
```

```
Out[ ]: array([[<AxesSubplot:title={'center': 'RNAI_seq_score'}>]], dtype=object)
```



```
In [ ]: df.hist(column="RNAP_seq_score")
```

```
Out[ ]: array([[<AxesSubplot:title={'center': 'RNAP_seq_score'}>]], dtype=object)
```



```
In [ ]: df["genus"].value_counts()
```

```
Out[ ]: Klebsiella      307
Escherichia      253
Salmonella       117
Enterobacter      61
Yersinia         38
Shigella         28
Citrobacter       23
Aeromonas        10
Raoultella        9
Erwinia           6
Pantoea           5
Serratia          4
Cronobacter       3
Pasteurella       2
Vibrio            2
```

```
Hafnia                2
Leclercia             2
Edwardsiella          2
Rahnella              1
Proteus               1
Pectobacterium        1
Enterobacteriaceae    1
Pseudomonas           1
Photobacterium        1
Name: genus, dtype: int64
```

```
In [ ]: df["family"].value_counts()
```

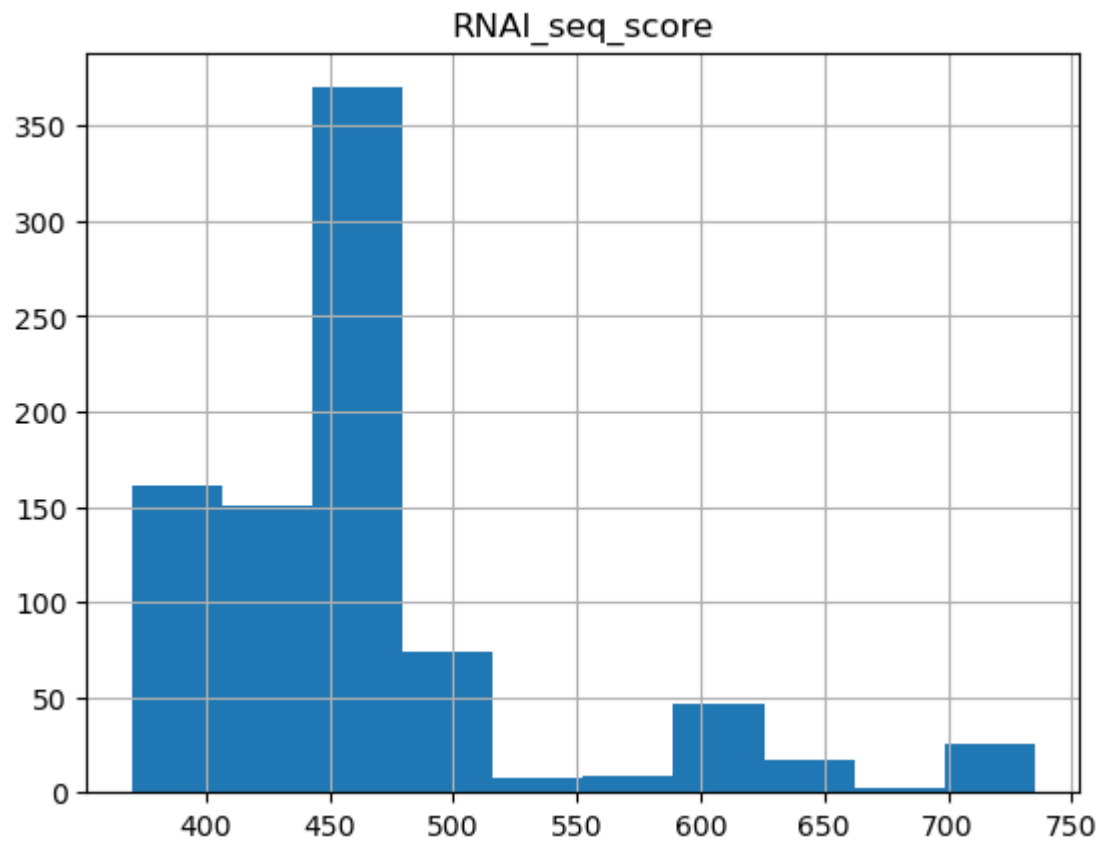
```
Out[ ]: Enterobacteriaceae    804
Yersiniaceae                 43
Erwiniaceae                  11
Aeromonadaceae              10
Hafniaceae                   4
Vibrionaceae                 3
Pasteurellaceae             2
Pseudomonadaceae            1
Morganellaceae              1
Pectobacteriaceae           1
Name: family, dtype: int64
```

```
In [ ]: df["order"].value_counts()
```

```
Out[ ]: Enterobacterales      864
Aeromonadales                10
Vibrionales                   3
Pasteurellales               2
Pseudomonadales              1
Name: order, dtype: int64
```

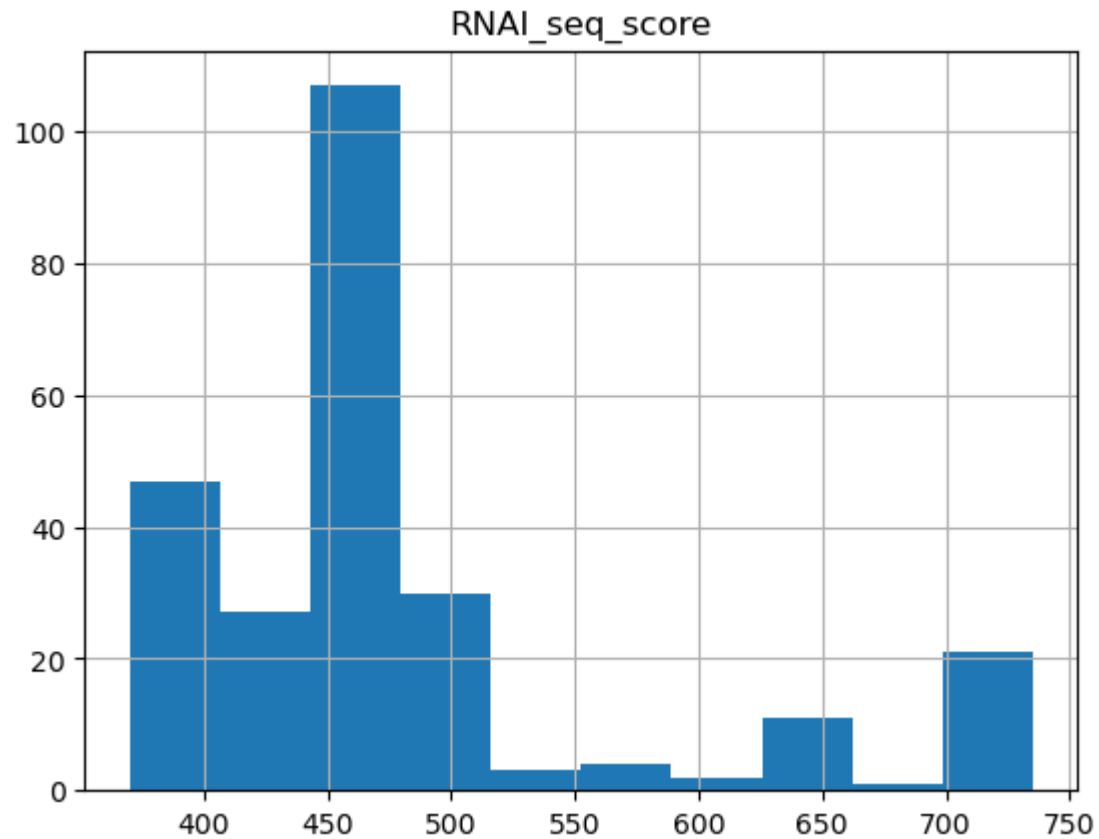
```
In [ ]: df[df["order"]=="Enterobacterales"].hist(column="RNAI_seq_score")
```

```
Out[ ]: array([[<AxesSubplot:title={'center': 'RNAI_seq_score'}>]], dtype=object)
```



```
In [ ]: df[df['genus']=="Escherichia"].hist(column='RNAI_seq_score')
```

```
Out[ ]: array([[<AxesSubplot:title={'center': 'RNAI_seq_score'}>]], dtype=object)
```



In []:

```
df[df["RNAI_seq_score"]>450].info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 492 entries, 11 to 879
Data columns (total 21 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                     492 non-null    object
1   species                               492 non-null    object
2   genus                                 492 non-null    object
3   family                                492 non-null    object
4   order                                 492 non-null    object
5   RNAI_promoter                         492 non-null    object
6   RNAI_promoter_direction               492 non-null    object
7   RNAP_promoter                         492 non-null    object
8   RNAP_promoter_direction               492 non-null    object
9   RNAI_seq_score                        492 non-null    int64
10  RNAI_seq_start                        492 non-null    int64
```

```
11 RNAI_seq_end          492 non-null  int64
12 RNAI_seq_alignmentA   492 non-null  object
13 RNAI_seq_alignmentB   492 non-null  object
14 RNAI_seq_direction    492 non-null  object
15 RNAP_seq_score        492 non-null  int64
16 RNAP_seq_start        492 non-null  int64
17 RNAP_seq_end          492 non-null  int64
18 RNAP_seq_alignmentA   492 non-null  object
19 RNAP_seq_alignmentB   492 non-null  object
20 RNAP_seq_direction    492 non-null  object
dtypes: int64(6), object(15)
memory usage: 84.6+ KB
```

```
In [ ]: df[df["genus"]=="Escherichia"].hist(column="RNAP_seq_score")
```

```
Out[ ]: array([[<AxesSubplot:title={ 'center': 'RNAP_seq_score' }>]], dtype=object)
```

